

Amirreza Sokhankhosh

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HIGHLIGHT OF RELEVANT SKILLS

- **Back-End Development:** Expertise in building and maintaining **RESTful APIs** using **Django**, **Nest.js**, **Express.js**, and **Flask**.
- **Front-End & Full-Stack Integration:** Hands-on experience aligning **React + Material UI** or **Tailwind** interfaces with backend services for seamless user experiences. Experience with using state management tools such as **MobX** and **React Query**.
- **Databases & Data Modeling:** Proficient with SQL and NoSQL systems (**PostgreSQL**, **MySQL**, **MongoDB**), including schema design, stored procedures, and optimized querying.
- **Programming Languages:** Strong foundation in **Python**, **TypeScript**, **JavaScript**, and **Java**.
- **Cloud & DevOps:** Hands-on with **AWS**, **Docker**, **GitHub Actions**, and CI/CD pipelines to ensure reliable and scalable deployments.
- **AI & Data Integration:** Applied **LLMs**, **Gen AI**, **time-series forecasting**, and predictive modeling to enhance application functionality and intelligence.
- **Collaboration & Agile Practices:** Experienced in **Scrum/Agile** environments, sprint planning, code reviews, and cross-functional teamwork.

EXPERIENCES

University of Manitoba

Graduate Research Assistant

Sep 2023 – Jul 2025

Winnipeg, Canada

- Developed and deployed **distributed backend systems** with **Flask**, **Express.js**, and **Docker**, enabling reliable orchestration of up to **36** computing nodes.
- Engineered a high-performance **REST API layer** to manage parallel training jobs and inter-node communication, achieving **90–100% system uptime**.
- Implemented modular, containerized **microservices** for task scheduling, validation, and monitoring, improving scalability and code maintainability.
- Optimized server workloads by offloading computation and streamlining inter-process communication, reducing execution time from **hours to minutes**.
- Collaborated on system design, benchmarking, and deployment pipelines; contributed to multiple **IEEE publications** on distributed systems engineering.

Bobo App Ltd.

Full Stack Developer Intern

May 2024 – Aug 2024

Winnipeg, Canada

- Faced the challenge of improving customer support efficiency; collaborated with a team at **MILA** to develop an **intelligent chatbot** that automated routine customer interactions, reducing response times.
- Accelerated backend development by designing and implementing **RESTful APIs** with **Supabase** and **PostgreSQL**, improving data retrieval performance and supporting scalable product features.
- Streamlined data management by automating CSV-to-SQL integration with Python scripts, reducing manual database population time by **80%**.
- Ensured smooth **full-stack integration** through cross-functional collaboration with the front-end team, aligning API contracts with user interface requirements to enhance product reliability.
- Enhanced team productivity within an **agile workflow** by using **Jira** and **Confluence** for task tracking, documentation, and sprint coordination.

Nadin Soft (Sadr Group Company)

Jul 2020 – Dec 2020

Full-stack Developer Intern

- Drove the successful completion of a critical internal project, addressing key deficiencies in API security, authentication, and data modeling for production deployment.
- Developed robust RESTful APIs with **Express.js**, implementing **JWT authentication** to secure user access and enable core application functionalities.
- Designed and implemented highly accurate MySQL database schemas for precise tracking and management of employee working and non-working hours, foundational for the project's business logic.
- **Deployed** fully functional features to the **live production server**.
- Contributed to an Agile Scrum development team, ensuring efficient progress and clear communication within sprints and **CI/CD pipelines**.

Aria

Sep 2025 – Oct 2025

Personal Project | [Code](#)

- Built a **full-stack platform** for generating tailored resumes and cover letters using **React** on the frontend and **Flask** on the backend.
- Developed a **modular backend** with clean REST endpoints, database indexing for faster queries, and applied **software design patterns** to improve code quality.
- Implemented a **caching layer** in PostgreSQL to optimize repeated requests, reducing the latency of API calls.
- Created a **responsive React frontend** with reusable components, **Material UI**, and secure authentication.
- Integrated **AI features** as a service via LangChain and LangGraph workflows, ensuring scalable, reliable generation pipelines without compromising system performance.

MarkMate

Jun 2025 – Aug 2025

Personal Project | [Code](#)

- Built a **full-stack AI grading platform** using **React + Tailwind** for the frontend and **Django** for the backend.
- Developed a **modular backend** with **Flask microservices**, JWT authentication, and clean REST APIs for reliable communication with AI services.
- Created a **responsive React frontend** with reusable components, **Context Provider** for state management, and **custom hooks** to streamline workflow interactions.
- Orchestrated **LangChain + LangGraph workflows** for AI grading, enabling scalable, maintainable, and deterministic evaluation pipelines.
- Implemented a **RAG pipeline** to retrieve course materials and improve grading quality, reducing manual review and ensuring consistent results.

EDUCATION

University of Manitoba

Sep 2023 – Aug 2025

Master of Science in Computer Science (GPA: 4.4 / 4.5)

Winnipeg, Canada

- **Relevant Coursework:** Security & Privacy, Deep Generative Modeling, Blockchain & Distributed Systems: A+

K.N. Toosi University of Technology

Sep 2018 – Feb 2023

Bachelor of Science in Computer Engineering

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, Node.js.

Front-end Frameworks & Tools: React, Material UI, Tailwind, React Query, MobX

Back-end Frameworks & Tools: Django, Express.js, NestJS, Flask

Databases: PostgreSQL, MongoDB, MySQL, Redis, Db2, Supabase.

Cloud & DevOps: Docker, Git, GitHub, Linux, CI/CD, AWS.

Tools & Methodologies: Jira, Confluence, Swagger, ClickUp, Agile, Scrum.

AI / Machine Learning: TensorFlow, PyTorch, Pandas, Transformers, LangChain.